

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	D.Sravanthi	Reg. No.:	192425315
Section / Batch:	AI & ML	Date:	2.08.2026

Code of Conduct

I D.Sravanthi(192425315) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D.Sravanthi

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3

Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, wordclass changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Implementation:

```
# Morphological Processing - Search Engine Indexing

words = ["analyzing", "analysis", "analytical"]

for word in words:
    if word.endswith("ing"):
        root = "analyze"
        structure = "analyze + ing"
        form = "Inflectional"
    elif word == "analysis":
        root = "analyze"
        structure = "analyze -> analysis"
        form = "Derivational"
    elif word == "analytical":
        root = "analyze"
        structure = "analysis + al"
        form = "Derivational"

    print("Word:", word)
    print("Structure:", structure)
    print("Type:", form)
    print("Normalized Root:", root)
    print()
```

Output:

```
*** Word: analyzing
    Structure: analyze + ing
    Type: Inflectional
    Normalized Root: analyze

    Word: analysis
    Structure: analyze -> analysis
    Type: Derivational
    Normalized Root: analyze

    Word: analytical
    Structure: analysis + al
    Type: Derivational
    Normalized Root: analyze
```

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

Implementation:

```
# Morphological Parsing - Sentiment Analysis

words = ["disagree", "agreement", "agreeable"]

for word in words:
    if word.startswith("dis"):
        root = "agree"
        structure = "dis + agree"
        change = "Prefix (Negation)"
    elif word.endswith("ment"):
        root = "agree"
        structure = "agree + ment"
        change = "Derivational (Noun)"
    elif word.endswith("able"):
        root = "agree"
        structure = "agree + able"
        change = "Derivational (Adjective)"

    print("Word:", word)
    print("Structure:", structure)
    print("Transformation:", change)
    print("Normalized Root:", root)
    print()
```

Output:

```
*** Word: disagree
    Structure: dis + agree
    Transformation: Prefix (Negation)
    Normalized Root: agree

Word: agreement
Structure: agree + ment
Transformation: Derivational (Noun)
Normalized Root: agree

Word: agreeable
Structure: agree + able
Transformation: Derivational (Adjective)
Normalized Root: agree
```

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

Implementation:

```
# Morphological Normalization - News Aggregation

words = ["govern", "government", "governance"]

for word in words:
    if word == "govern":
        root = "govern"
        structure = "govern"
        word_class = "Verb"
    elif word.endswith("ment"):
        root = "govern"
        structure = "govern + ment"
        word_class = "Noun"
    elif word.endswith("ance"):
        root = "govern"
        structure = "govern + ance"
        word_class = "Noun"

    print("Word:", word)
    print("Structure:", structure)
    print("Word Class:", word_class)
    print("Normalized Root:", root)
    print()
```

Output:

```
Word: govern
Structure: govern
Word Class: Verb
Normalized Root: govern

Word: government
Structure: govern + ment
Word Class: Noun
Normalized Root: govern

Word: governance
Structure: govern + ance
Word Class: Noun
Normalized Root: govern
```

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

Implementation:

```
# Morphological Parsing - Document Classification

words = ["activate", "activation", "reactivation"]

for word in words:
    if word == "activate":
        root = "activate"
        structure = "activate"
        meaning = "Base Verb"

    elif word == "activation":
        root = "activate"
        structure = "activate + ion"
        meaning = "Verb -> Noun"

    elif word == "reactivation":
        root = "activate"
        structure = "re + activate + ion"
        meaning = "Repeated Action"

    print("Word:", word)
    print("Structure:", structure)
    print("Meaning:", meaning)
    print("Normalized Root:", root)
    print()
```

Output:

```
Word: activate
Structure: activate
Meaning: Base Verb
Normalized Root: activate

Word: activation
Structure: activate + ion
Meaning: Verb -> Noun
Normalized Root: activate

Word: reactivation
Structure: re + activate + ion
Meaning: Repeated Action
Normalized Root: activate
```

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

Implementation:

```
# Morphological Normalization - Search Optimization

words = ["create", "creates", "creating"]

for word in words:
    if word == "create":
        root = "create"
        structure = "create"
        transform = "Base Form"

    elif word.endswith("s"):
        root = "create"
        structure = "create + s"
        transform = "Inflectional (3rd Person Singular)"

    elif word.endswith("ing"):
        root = "create"
        structure = "create + ing"
        transform = "Inflectional (Present Participle)"

    print("Word:", word)
    print("Structure:", structure)
    print("Transformation:", transform)
    print("Normalized Root:", root)
    print()
```

Output:

```
*** Word: create
    Structure: create
    Transformation: Base Form
    Normalized Root: create

Word: creates
Structure: create + s
Transformation: Inflectional (3rd Person Singular)
Normalized Root: create

Word: creating
Structure: create + ing
Transformation: Inflectional (Present Participle)
Normalized Root: create
```

Rubrics for Calculation

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Criteria		Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)		Level 2 (Developing)	Level 1 (Beginning)	
Understanding of Scenario		20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps		Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues		20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions		Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts		25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors		Limited or partially correct application	Unable to apply relevant concepts	
Decision Making		20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification		Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response		15%	Response is clear, structured, and logically presented	Generally clear with minor issues		Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Q. No. / Criteria Level	Understanding of Scenario		Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1								

2						
3						
4						
5						